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[22CSE110]

THIRD SEMESTER B.Tech EXAMINATION, OCTOBER-2023

DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ii. ANSWER ALL QUESTIONS

		PART-A			**05 X 02=10**
1.	**a.**	Define: **i.** Atomic attribute **ii.** Single valued attribute.	L-1	CO1	**02**
	b.	Define Referential integrity constraint.	L-1	CO2	**02**
	c.	Mention the different data types used in SQL.	L-1	CO3	**02**
	d.	Explain View in SQL.	L-1	CO4	**02**
	e.	Define Third Normal Form.	L-1	CO5	**02**
		PART-B			**06 X 05=30**
2.	**a.**	Define schema, Describe 3-schema architecture of DBMS with logical and physical data independence.	L-2	CO1	**06**
		OR			
	b.	Explain the various DBMS interfaces.	L-2	CO1	**06**
3.	**a.**	With respect to ER model explain with example **i.** Composite attributes **ii.** Cardinality Ratio **iii.** Binary relationship	L-2	CO1	**06**
		OR			
	b.	Design an ER diagram for a hospital management system. Assume suitable entity types, attributes and relations (Minimum 3 entities).	L-3	CO1	**06**
4.	**a.**	List and explain the characteristics of relations.	L-2	CO2	**06**
		OR			
	b.	Briefly discuss how update operations on a relation deal with constraint violations.	L-2	CO2	**06**
5.	**a.**	Explain INSERT, DELETE and UPDATE command in SQL with example.	L-2	CO3	**06**
		OR			

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	b.	Explain specifying constraints in SQL.	L-2	CO3	06
6.	a.	How a view is created and dropped? What problems are associated in updating the views?	L-2	CO4	06
		OR			
	b.	Explain triggers in SQL with example.	L-2	CO4	06
7.	a.	Explain Transaction support in SQL.	L-2	CO5	06
		OR			
	b.	Explain 1NF, 2NF and 3NF with examples.	L-2	CO5	06
		PART-C		**6 X 10=60**	
8.	a.	Discuss the advantages of DBMS by comparing with file systems.	L-2	CO1	10
		OR			
	b.	Explain the component modules of DBMS.	L-2	CO1	10
9.	a.	Explain the steps involved in ER-Relational mapping.	L-2	CO2	10
		OR			
	b.	Explain the SELECT and PROJECT operations in relational algebra with suitable examples.	L-2	CO2	10
10.	a.	Explain all the aggregate functions in SQL.	L-2	CO3	10
		OR			
	b.	Explain all the DDL commands with syntax and example	L-2	CO3	10
11.	a.	Write SQL queries for the following database schema: STUDENT(Usn, Name, Branch, Percentage) FACULTY(Fid, Fname, Department, Designation , Salary) COURSE(Cid, Cname, Fid) ENROLL(Cid, Usn, Grade) **i.** Retrieve the names of all students enrolled for the course CSE110 **ii.** List all the departments having an average salary of all faculties above Rs 10,000. **iii.** Find the fid with second highest salary) **iv.** List the names of students enrolled for the course CSE110 and having A grade.	L-2	CO4	10
		OR			

	b.	Write SQL queries for the following database schema: HOTEL(Hotelno, name , address) ROOM(Roomno, Hotelno, type, price) BOOKING(Hotelno, Guestno, datefrom, dateto, Roomno) GUEST(Guestno, name, address) **i.** List the names and addresses of all guest booked the hotel located in Bangalore ordered by name. **ii.** List all the family rooms with a price below Rs 400. **iii.** How many hotels are there? **iv.** List the number of rooms.	L-2	CO4	**10**
12.	**a.**	Explain the informal design guidelines for relational schemas.	L-2	CO5	**10**
		OR			
	b.	Explain 4NF and 5NF with examples.	L-2	CO5	**10**
13.	**a.**	Explain the desirable properties of Transaction.	L-2	CO5	**10**
		OR			
	b.	Explain Conflict and View Serializability in Transaction.	L-2	CO5	**10**